

Welcome to the ML Course

Machine Learning for Science & Discovery

MSU REU 2026

About Me

Prof. Danny Caballero

- Physics Education Research (PER) background (PhD, Georgia Tech)
- Research in STEM education, computational modeling, and machine learning
- At MSU PA since 2013, joined CMSE in 2021
- Worked with machine learning in research and teaching
- Enjoy teaching ML using real data, asking real questions, solving real problems

This class is about **doing ML like a researcher does**: asking questions, building intuition, learning from failure.

What This Class Looks Like

Format

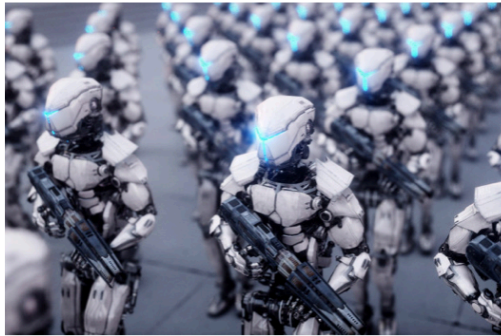
- **Slides + Activities:** Theory and framing in slides, hands-on coding in Jupyter notebooks
- **Real datasets:** Not toy data, but working with research problems that you'll encounter
- **Work in pairs/groups:** You learn better together; science is collaborative
- **Ask questions:** No such thing as a dumb question. If you're confused, others are too.

What We'll Build

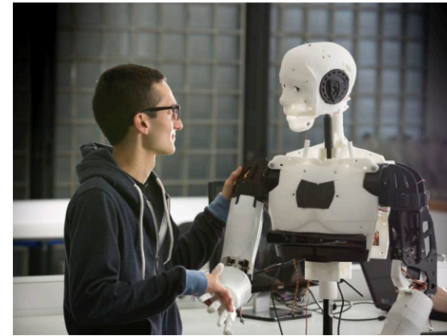
By the end of the class, you'll have built **classifiers, regressors, and model pipelines** on a real dataset.

What is Machine Learning?

Machine Learning



What society thinks we do



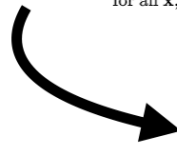
What our friends think we do



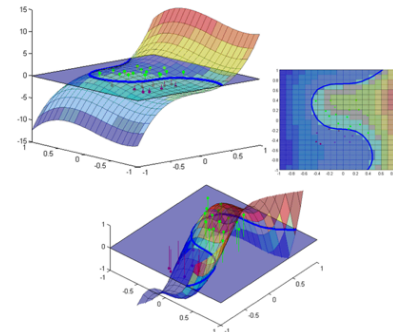
What our families thinks we do

$$K_k^{cc}(\mathbf{x}, \mathbf{z}) = \sum_{k_1, \dots, k_d, \sum_{j=1}^d k_j = k} \frac{k!}{k_1! \dots k_d!} \left(\frac{1}{d}\right)^k \prod_{j=1}^d 1_{[2^{k_j} x_j] = [2^{k_j} z_j]},$$

for all $\mathbf{x}, \mathbf{z} \in [0, 1]^d$.



What my boss thinks we do



What we think we do

```
from sklearn import linear_model as lm

X = iris[["petal_length"]]
y = iris["petal_width"]

# Fit the linear model
model = lm.LinearRegression()
results = model.fit(X, y)

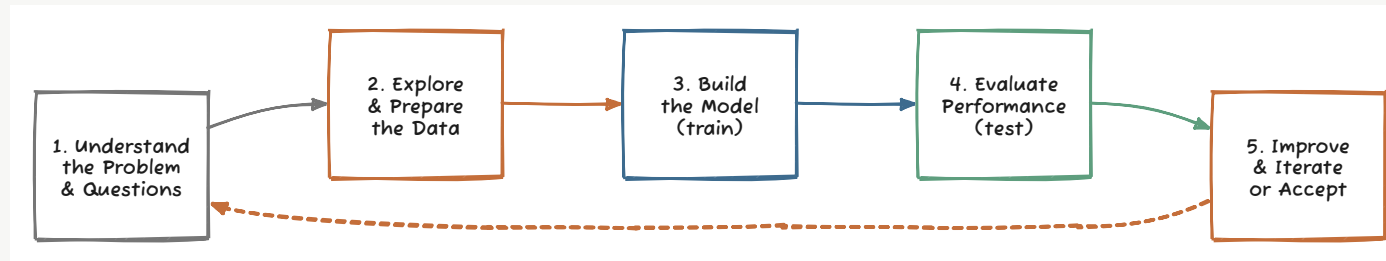
# Print the coefficients
print model.intercept_, model.coef_

-0.363075521319 [ 0.41575542]
```

What we actually do

Supervised Learning as a Framework

Not a magic algorithm. A systematic process you'll repeat.



1. **Understand** — Ask a real question about your data
2. **Explore** — Clean data, spot patterns, prepare for modeling
3. **Build** — Train a model on labeled examples
4. **Evaluate** — Test on held-back data. Does it work?
5. **Improve** — Refine or try a different approach (↺ loop back)

Learning by Doing

You won't just see code—you'll **write it, break it, fix it, understand it**.

This is messy. That's the point.

- Day 1: **Load real data** (it's broken, we'll find it)
- Days 2–3: **Build models** (classification, regression, tuning)
- Day 4: **Put it together** (real workflow, start to finish)

By the end, you'll have **intuition** about how Supervised ML actually works.

A Note on Struggle

Machine learning is hard. So is programming.

This is normal. Scientists struggle with new tools all the time. What matters is:

- You try
- You read documentation
- You ask for help
- You learn from what breaks

I'm here to help you get unstuck, and I don't have all the answers.

Let's Start With Introductions

Tell us:

- Your name and where you're from
- One thing you're curious about (in ML, science, or anything)
- One thing you're excited to learn in this class

We'll go around the room. Keep it short—30 seconds each.

Today's Goal

By the end of today, you'll:

1. Load a real dataset (100,000 astronomical objects)
2. Spot problems in the data before analysis
3. Visualize relationships between features
4. Understand what makes a good question to ask with data